

CODE SPORT



In this month's column, we discuss a medley of computer science interview questions.

In last month's column, we had discussed a number of questions. A few readers did send in their solutions and answers to those questions.

I will discuss those answers in next month's column, but would like others too to send in their solutions for discussion. Meanwhile, in this month's column, we will continue to discuss some more computer science interview questions in algorithms, machine learning and natural language processing.

1. We are all familiar with spoken language recognition systems, which can perform speech to text (STT) recognition such as Apple's 'Siri', 'Google Now' and Microsoft's 'Cortona'. These systems have to deal with a number of complexities as different speakers have widely varying accents and use different styles of pronunciation. Let us consider a person who is reciting the poem 'Twinkle, twinkle, little star; How I wonder what you are'. Let us assume that the STT system so far has correctly identified all the words spoken until the last word 'are'. Now it needs to identify the last word as 'are'. You are given access to a large corpus of English language on which the STT system can be trained. How will you design an algorithm that can recognise the poem completely, under the assumption that it has been able to correctly recognise all previous words in that poem spoken so far?
2. Most of you must be familiar with language models in NLP. These models are used to predict the most probable next word from a given vocabulary, given a sequence of words that has already been seen. One of the simplest ways of predicting the next word is to use count based approaches. For example, if we have the unigram count of a given corpus, we can simply use the word frequency to predict the first word in the sentence, picking up the word with the highest frequency. Once we have predicted the first word and we have the bigram counts of a given corpus, we can extend it with the most frequent bigram containing the first word, and so on. What will you do if the given sentence contains words that are not present in the vocabulary and you are asked to predict the next word?
3. Let us again consider the STT system mentioned in question (1) above. We are given a spoken sentence in English and have been asked to convert it to text. For example, consider the sentence: "I took a flight to New York." Let us assume that the phrase 'New York' did not occur at all in the training corpus you have used to train the STT system. However, the training corpus did contain a sentence: "I took a flight to Paris." Can you explain how your language model will recognise the phrase 'New York'?
4. A language model is useful in building a number of NLP applications. One of the possible uses is the auto-completion of a query or auto correction of a mistyped query. Can you explain how a language model can be used for either of these two tasks?
5. Consider the question above. Language models are typically trained on formal sentences that are syntactically/grammatically correct. However, when users type queries on search engines, they are informal, just a bag of words and may not be grammatically correct. Will this cause an issue in using a language model to auto-complete the query?
6. You have a corpus of documents in multiple languages. You want to provide search engine capability across this document corpus. For supporting the auto-completion feature, will you train different language models for each language supported? Else, how will you support the auto-completion feature?